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Regression Analysis Project Report: Modeling Building Energy Consumption

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I. Executive Summary:

a. To support Lux Inc.'s energy-efficiency initiative across the region, this project aims to develop a predictive regression model for daily energy consumption. By analyzing the relationship between distinctive characteristics like building type, square footage, number of occupants, number of appliances used, among others, the model will provide personalized efficiency recommendations and inform resource allocation planning. The analysis is based on a dataset of 80 observations representing daily energy consumption across different buildings in the city. The dataset includes numerical variables such as Square Footage, Number of Occupants, Appliances Used, and Average Temperature, as well as categorical variables such as Building Type and Day of the Week. The exploratory analysis revealed that square footage has the strongest positive linear relationship with energy consumption, making it the most influential. On the other hand, average temperature shows a slight negative trend, suggesting that heating demands on colder days use more consumption than cooling needs.

b. To initiate the evaluation of the factors that influence energy consumption, the analysis began with a progressive modeling approach. Model 1 focused solely on the structural and physical characteristics of the buildings, including square footage, number of occupants, appliances used, and building type as predictors. Model 2 incorporated the Average Temperature to capture environmental effects. The regressors appear to be significant in this model, however the Residuals Versus Fits Plot appeared to be divided in 2 groups. To optimize model fit, a quadratic term (Average Temperature^2) was included in Model 3. Using backward elimination, non-significant interaction terms were systematically removed. Model 4 adds Day of the Week as a categorial variable and reflects distinctive operational patterns between weekdays and weekends. In Model 5, interaction terms were introduced to improve model fit and enhance the behavior of residual diagnostics. Specifically, two interactions were added: Square Footage x Average Temperature and Average Temperature x Building Types. These interactions were included to capture how temperature effects vary depending on building size and type. This resulted in noticeable improvements in the Normal Probability Plot.

II. Introduction:

a. As technological advancements accelerate, as well as the demand for mobile devices, energy has become essential for everyday life. That explains why the US Energy Information Administration reported that buildings accounted for 36.9% of the total energy consumption in the US (2023). In response to this trend, the utility company Lux Inc started compiling data to better understand the factors that influence energy consumption.

b. The data collected contains data from random buildings, and is composed of the following: building type, square footage, number of occupants, appliances used, average temperature, and day of the week. With the following data, a modelling process will take place starting with the structural variables- building type, square footage, number of occupants, appliances used- to evaluate their relationship with energy consumption. Later, temperature and days of the week will be added to see their effect on the previous structures. The goal of this model is to predict daily energy consumption to enable personalized efficiency recommendations and more effective resource allocation.

III. Data Description

a. **Description of dataset and variables:**

i. **Table 1.** Summary Statistics for Numerical Variables

1. The dataset contains information from multiple buildings across the city with the purpose of understanding the factors that influence daily energy consumption (kWh). The dataset includes a combination of quantitative and qualitative predictors that are relevant to building energy consumption patterns.

2. The variables included are:

Variable	Type	Description
Energy Consumption (kWh)	Response (numeric)	Daily energy usage of each building.
Square Footage	Numeric	Total floor area of the building in square feet.
Number of Occupants	Numeric	Number of people regularly present in the building.
Appliances Used	Numeric	Count of major appliances such as AC units or refrigerators.
Average Temperature (°C)	Numeric	Average outdoor temperature on the recorded day.
Building Type	Categorical	Classification of the building (Residential, Commercial,

		Industrial).
Day of Week	Categorical	Weekday vs. Weekend.

b. Summary statistics table:

i. Notes on any data cleaning or transformations performed:

1. No missing values or inconsistencies were identified
2. No transformations (log, sqrt, standardization) were necessary at this stage.
3. Categorical variables (Building Type, Day of Week) were dummy coded by Minitab using the (1,0) reference coding system.
4. All numerical variables were kept on their original scale.

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Energy Consumption	80	0	4212.0	95.3	852.0	2352.0	3606.6	4289.2	4915.7	6042.6
Square Footage	80	0	26293	1521	13605	1306	14047	27374	39134	49354
Number of Occupants	80	0	45.98	3.39	30.35	2.00	20.25	41.50	73.00	99.00
Appliances Used	80	0	28.76	1.54	13.82	1.00	17.25	28.00	41.75	49.00
Average Temperature	80	0	21.791	0.795	7.109	10.400	15.127	21.970	27.540	34.710

ii. **Figure 1.** Descriptive Statistics Output

IV. Exploratory Data Analysis (EDA)

a. **Description of dataset and variables:**

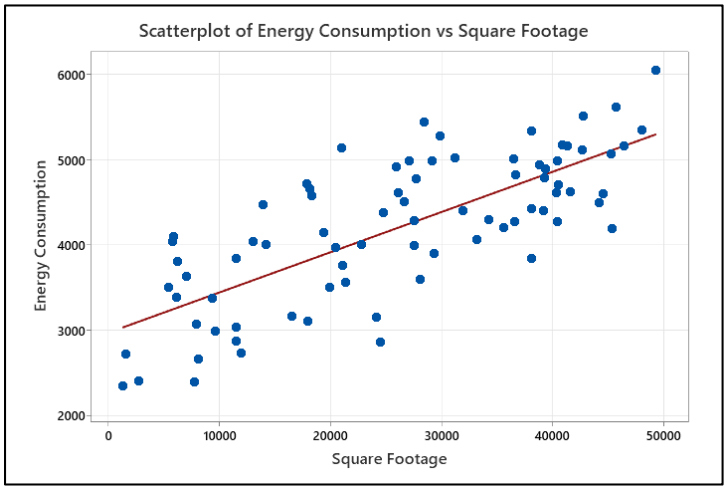
i. The dataset contains 80 observations representing daily energy consumption across a variety of buildings with different structural, occupancy, and operational characteristics. It includes both numerical and categorical variables that capture relevant drivers of energy usage, such as building size, number of occupants, appliance load, and environmental temperature. A detailed numerical summary of all quantitative variables is provided in **Table 1**, which shows the central tendency and dispersion of key predictors used in the regression analysis.

b. Summary of descriptive statistics:

i. **Scatterplots of each regressor with the response variable:**

1. To visually explore the relationships between the independent variables and the response variable (*Energy Consumption*), individual scatterplots were generated for each numerical predictor. These graphical tools

allow for an initial assessment of linearity, potential trends, variability, and the presence of unusual observations that could impact model performance. The following subsections summarize the main patterns observed in the scatterplots and provide insights regarding the strength and direction of each relationship.



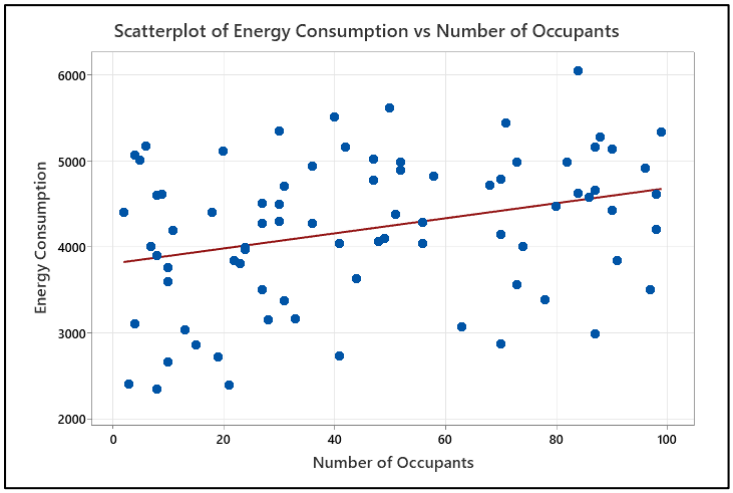
a. **Figure 2-** Scatterplot: Energy Consumption vs. Square Footage

   i. Positive linear relationships.

   ii. Larger buildings tend to consume more energy.

   iii. Strongest visual pattern among predictors.

   iv. Justifies including Square Footage in all candidate models.



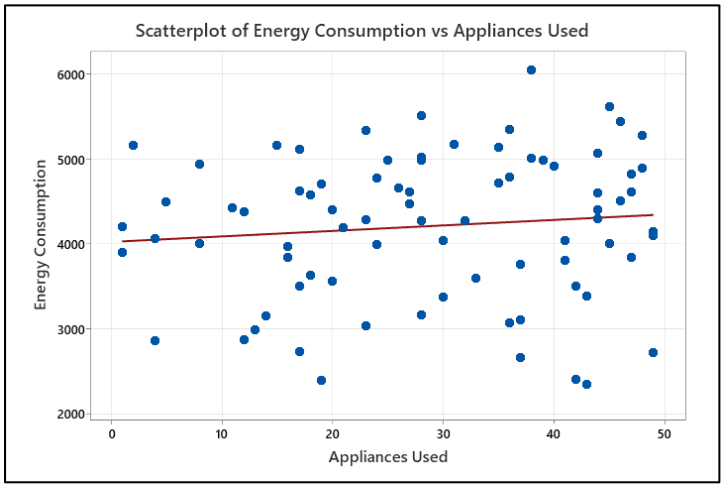
b. **Figure 3-** Scatterplot: Energy Consumption vs. Number of

Occupants

i. Slight positive trend.

ii. Relationships are weaker and more dispersed compared to square footage.

iii. Occupancy still has a logical influence on energy usage and should be considered.

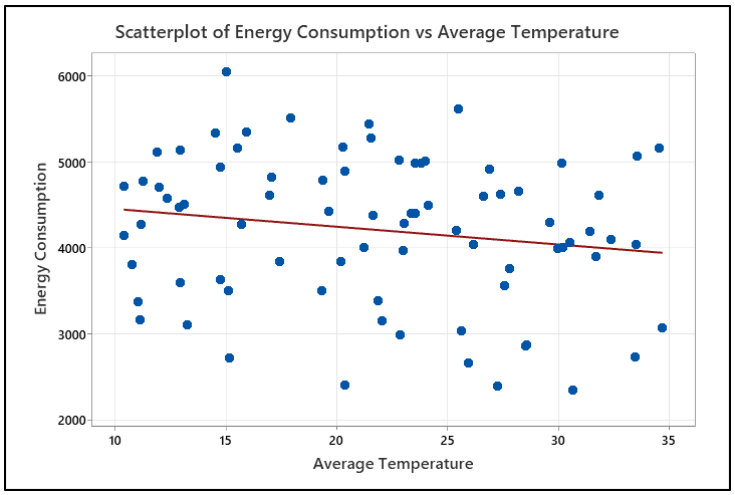


c. **Figure 4 –** Scatterplot: Energy Consumption vs. Appliances Used

i. Very weak positive trend; data points are highly scattered.

ii. Appliances contribute to usage but may not be strongly predictive on their own.

iii. Possible interaction with Building Type.

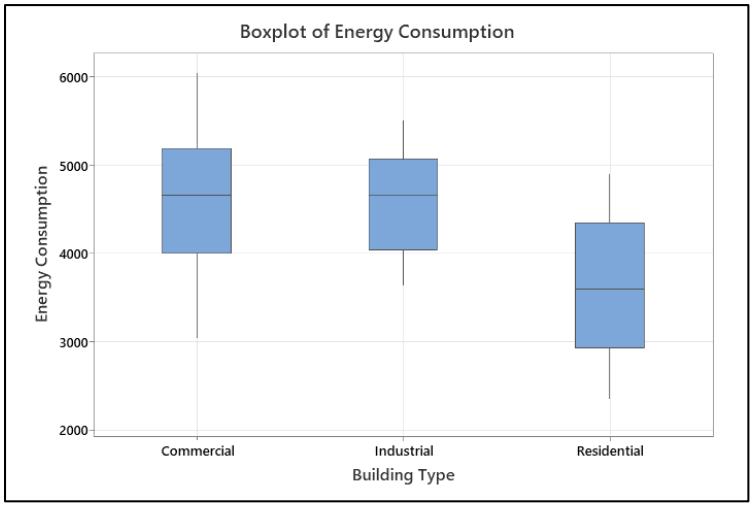


d. **Figure 5** - Scatterplot: Energy Consumption vs. Average Temperature

   i. Slight negative trend: as temperature increases, energy consumption tends to slightly decrease.

   ii. This makes sense if heating (cold days) consumes more energy than cooling.

   iii. Suggests a temperature-sensitive model.



e. **Figure 6** - Boxplot of Energy Consumption by Building Type

   i. Industrial buildings have the highest energy consumption, with a much higher median.

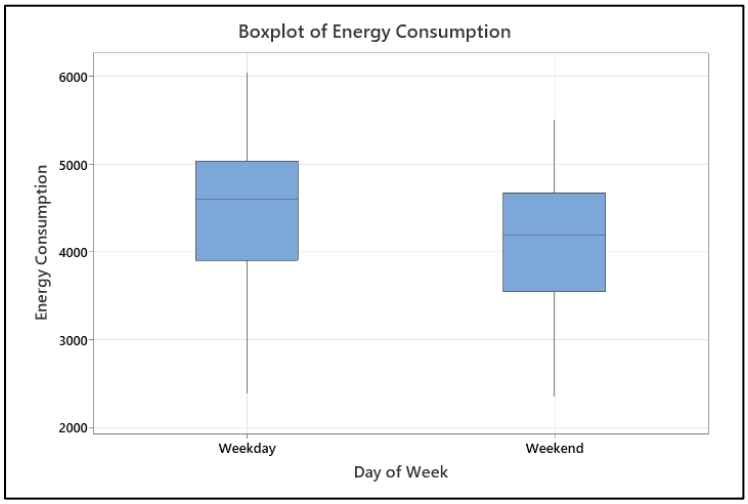
   ii. Residential buildings show the lowest consumption,

reflecting lower operational demands.

iii. Commercial buildings fall in the middle, showing moderate energy usage.

iv. Industrial buildings also show the largest variability, indicating differing sizes or operational loads.

v. Clear separation between the three groups suggests that Building Type strongly influences energy consumption.



f. **Figure 7** - Boxplot of Energy Consumption by Day of Week

i. Weekdays show higher energy consumption on average compared to weekends.

ii. The median consumption is noticeably higher during weekdays.

iii. The spread (IQR) is wider for weekdays, indicating more variability in energy usage.

iv. Weekends show a lower overall distribution, suggesting reduced building activity.

v. No extreme outliers appear, and both groups show stable patterns.

c. **Observations and insights from data visualization:**

i. Square Footage has the strongest linear association with Energy Consumption.

ii. Occupants and Appliances Used show mild positive relationships.

iii. Average Temperature exhibits a small but consistent negative slope.

iv. Building Type and Day of Week likely explain additional categorical variation, especially because commercial and industrial buildings generally show higher consumption.

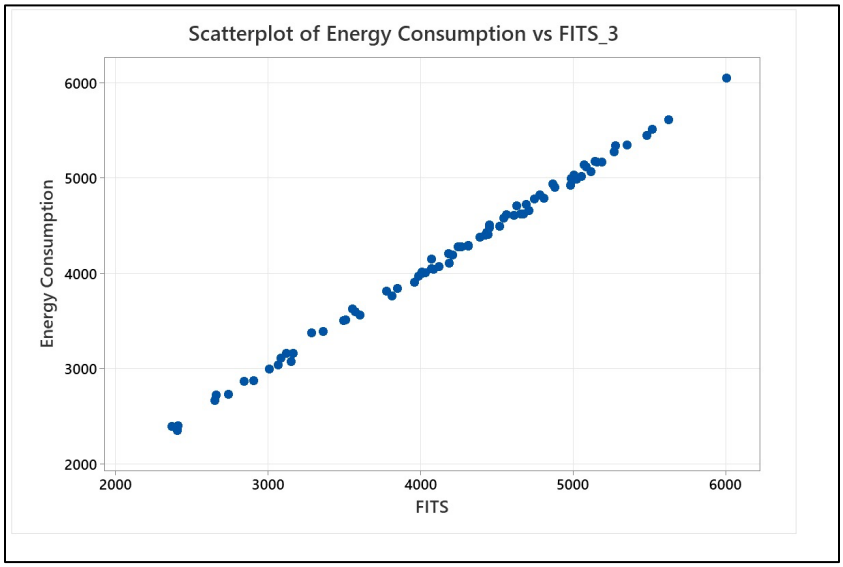
v. No strong nonlinear patterns were observed; linear models appear appropriate.

vi. No major outliers were evident visually, except for one detected later in diagnostics (Observation 53).

V. Regression Modeling and Results:

a. Regression Model 1:

i. Regression Model Fitting of Model 1:



1. **Figure 8** – Regression Model Fitting of Model 1

ii. Regression Model Equation:

1. The Regression Model 1 equation is:

   a. $\hat{Y} = \beta_0 + \beta_1$*Square Footage + β_2*Number of Occupants + β_3*Appliances Used

Regression Equation

Building Type			
Commercial	Energy Consumption	=	1878.0 + 0.050434 Square Footage + 10.109 Number of Occupants + 20.702 Appliances Used
Industrial	Energy Consumption	=	2378.0 + 0.050434 Square Footage + 10.109 Number of Occupants + 20.702 Appliances Used
Residential	Energy Consumption	=	1372.4 + 0.050434 Square Footage + 10.109 Number of Occupants + 20.702 Appliances Used

b. **Figure 9**– Regression Model Equation of Model 1

2. The regression model presented three equations that modeled energy consumption per building type, using only variables that affected the building's physical load as predictors. It is imperative to know the baseline energy demand for each building type. Clearly, building type affects the baseline energy levels given by the intercept in each model. Industrial buildings had the highest base load (Intercept = 2378.0 kW/H), followed by commercial buildings (Intercept = 1878.0 kW/H), and lastly residential buildings (Intercept = 1372.4 kW/H). Regardless of other variables, Industrial buildings require a higher energy base load. Similarly, the physical variables (the regression coefficients) impacted each regression equation. The most impactful physical variable was Appliances Used; its coefficient yielded 20.702 kW/H per appliance used in each building type. On the other hand, each additional occupant contributed 10.109 kW/H to the overall regression model. The smallest coefficient was Square Footage; however small, the magnitude of each building type makes it significant.

iii. Analysis of Variance of Model 1:

1. The ANOVA Model aimed to confirm or reject the existence of a statistically significant relationship between the variables and energy consumption. Since the P-value of each coefficient is less than the critical value of 0.05, there exists a statistically significant relationship between the variables and energy consumption. Additionally, the Regression Sum of Squares accounted for almost all the Sum of Squares and consequently, almost all its variation.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	5	57231574	11446315	6978.33	0.000
Square Footage	1	32660676	32660676	19911.83	0.000
Number of Occupants	1	7408324	7408324	4516.54	0.000
Appliances Used	1	6002078	6002078	3659.21	0.000
Building Type	2	13993938	6996969	4265.76	0.000
Error	74	121380	1640		
Total	79	57352954			

2. **Figure 10 -** Analysis of Variance (ANOVA) for Model 1

iv. T-Tests of Model 1:

1. Furthering the analysis to determine if the variables were statistically significant or not, a T-test was performed. All variables remained in the model since they had a low P-value (less than 0.05). Square Footage had the highest T-value, this confirms that it is a statistically significant predictor and adds that it is the most dominant predictor for energy base load.

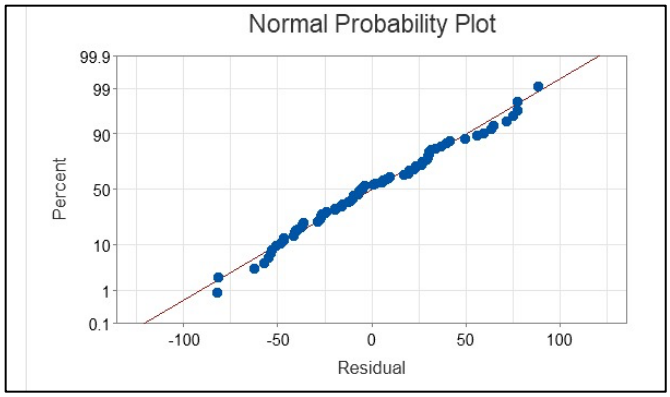
Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	1878.0	20.4	91.91	0.000	
Square Footage	0.050434	0.000357	141.11	0.000	1.14
Number of Occupants	10.109	0.150	67.21	0.000	1.00
Appliances Used	20.702	0.342	60.49	0.000	1.08
Building Type					
Industrial	500.0	12.4	40.37	0.000	1.61
Residential	-505.6	11.6	-43.56	0.000	1.59

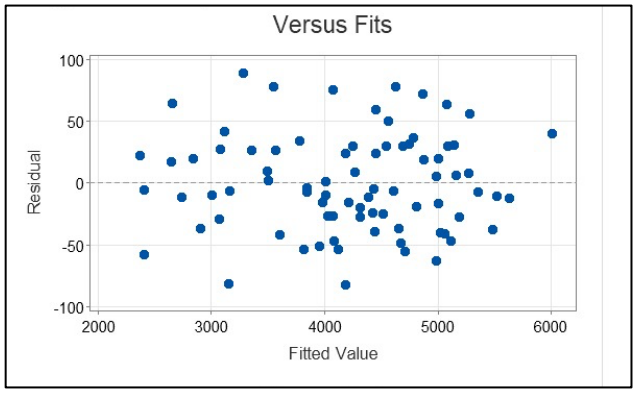
2. **Figure 11 –** T-Test for Model 1

v. Residual Analysis of Model 1(Including Plots and Discussions):

1. The Residual Analysis helped identify normality and constant variance in the model. The Normal Probability Plot aims to identify outliers that skew the data, and although small, the data follows a slight S-curve on its edges. The residuals within the model, mapped in the Normal Probability Plot, deviate slightly from the Normal Distribution. The scatterplot of Fitted Values vs. Fits of Model 1 slightly clumps the data between values 3000 and 5000 on the X-axis. However, it is not a cause for concern since it indicates that most buildings in the evaluated dataset are large consumers. On the other hand, that slight clump is not indicative of an absence of constant variance. Constant variance was still presumed, given that the vertical spread remains stable.



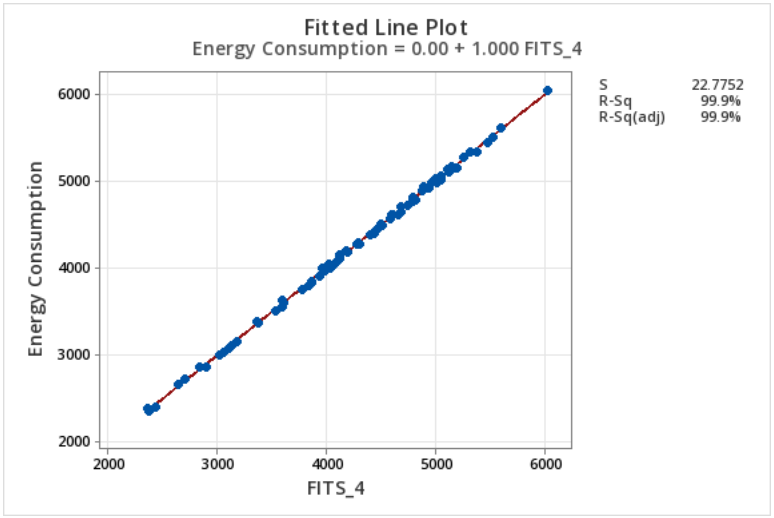
2. **Figure 12** – Normal Probability Plot of Model 1



3. **Figure 13 –** Scatterplot of Fitted Values vs. Fits of Model 1

b. Regression Model 2:

i. Regression Model Fitting of Model 2:

1. 

2. **Figure 12 –** Regression Model Fitting of Model 2

ii. Regression Model Equation:

1. The Regression Equation for Model 2 is:

   a. $\hat{Y} = \beta_0 + \beta_1$ (Square Footage) + β_2 (Number of Occupants) + β_3 (Appliances Used) + β_4 (Average Temperature) + β_5 (Building Type)

Regression Equation

Building Type			
Commercial	Energy Consumption	=	1994.0 + 0.050280 Square Footage + 10.0372 Number of Occupants + 20.280 Appliances Used - 4.446 Average Temperature
Industrial	Energy Consumption	=	2490.6 + 0.050280 Square Footage + 10.0372 Number of Occupants + 20.280 Appliances Used - 4.446 Average Temperature
Residential	Energy Consumption	=	1491.9 + 0.050280 Square Footage + 10.0372 Number of Occupants + 20.280 Appliances Used - 4.446 Average Temperature

1. **Figure 13 –** Regression Equation of Model 2

iii. Analysis of Variance of Model 2:

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	**6**	**57307192**	**9551199**	**15236.37**	**0.000**
Square Footage	1	32329944	32329944	51573.73	0.000
Number of Occupants	1	7266438	7266438	11591.65	0.000
Appliances Used	1	5576431	5576431	8895.70	0.000
Average Temperature	1	75618	75618	120.63	0.000
Building Type	2	13682008	6841004	10912.98	0.000
Error	**73**	**45761**	**627**		
Total	**79**	**57352954**			

1. **Figure 14 –** Analysis of Variance (ANOVA) of Model 2

2. In the Analysis of Variance, we use the p-value 0.05 as criteria to reject the hypothesis that the model is not significant. Since the p-value of the model was of 0.000, we reject the hypothesis, therefore Model 2 is significant.

iv. T-Tests of Model 2

1. For all the regressors, we use the p-value 0.05 as criteria to reject the hypothesis that the regressor is not significant. Since the p-value for all the regressor was of 0.000, we reject the hypothesis for all the regressors, meaning that all the regressors in Model 2 are significant.

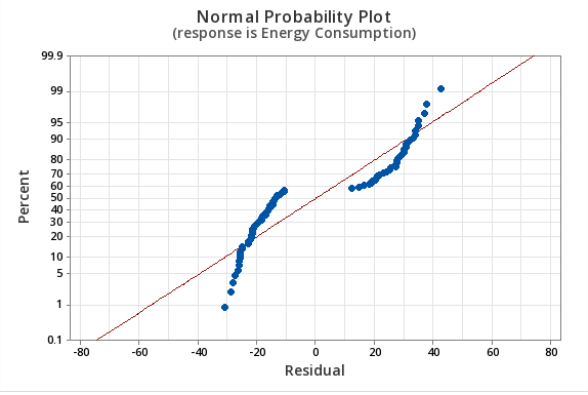
Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	1994.0	16.5	121.11	0.000	
Square Footage	0.050280	0.000221	227.10	0.000	1.14
Number of Occupants	10.0372	0.0932	107.66	0.000	1.01
Appliances Used	20.280	0.215	94.32	0.000	1.11
Average Temperature	-4.446	0.405	-10.98	0.000	1.04
Building Type					
Industrial	496.61	7.66	64.80	0.000	1.61
Residential	-502.11	7.18	-69.90	0.000	1.60

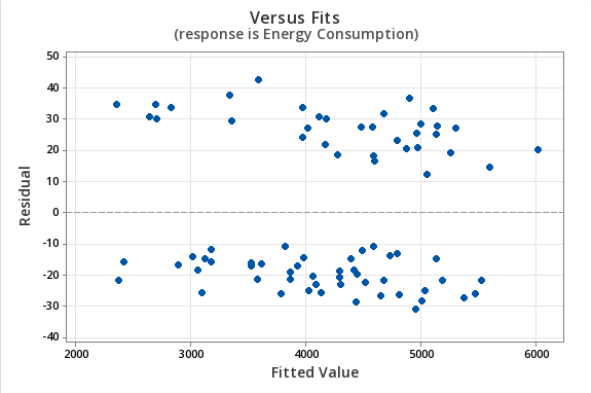
1. **Figure 15 –** T – Test of Model 2

v. Residual Analysis of Model 2 (Including Plots and Discussions):

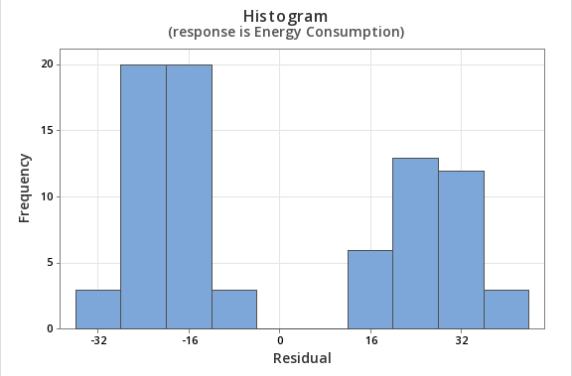
1. In all the residual plots, the residuals appear to be divided into two distinct groups. In the Normal Probability Plot, the Model 2 doesn't follow the regression, meaning that the residuals don't follow a Normal Distribution. However, in the Histograms it's observed that both groups individually do follow a normal distribution. Either this indicates that the model is mixing two structurally set of observations or that interactions of variables are missing in the variable. Considering that this division occurs when adding the temperature variable, the recommendation for the next model is to add interactions with these variables.



1. **Figure 16 –** Normal Probability Plot of Model 2

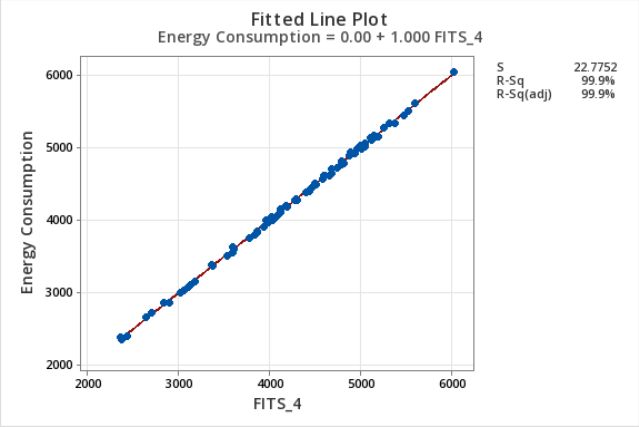


2. **Figure 17 –** Scatterplot of Fitted Values vs. Fits of Model 2



3. **Figure 18 –** Histogram Plot of Model 2

c. Regression Model 3:



i. **Figure 19 –** Regression Model Fitting of Model 3

ii. Regression Equation of Model 3:

1. Based on Model 2, the variable Temperature was squared (Temperature^2) and added to the model with all its interactions with the rest of the

variables. Later, regressors were deleted with backward elimination until obtaining this equation for Model 3:

Regression Equation

Building Type			
Commercial	Energy Consumption	=	2069.7 + 0.048986 Square Footage + 10.0253 Number of Occupants + 20.336 Appliances Used - 7.98 Average Temperature + 0.0000589 Square Footage*Average Temperature
Industrial	Energy Consumption	=	2513.7 + 0.048986 Square Footage + 10.0253 Number of Occupants + 20.336 Appliances Used - 5.528 Average Temperature + 0.0000589 Square Footage*Average Temperature
Residential	Energy Consumption	=	1502.9 + 0.048986 Square Footage + 10.0253 Number of Occupants + 20.336 Appliances Used - 5.009 Average Temperature + 0.0000589 Square Footage*Average Temperature

a. $\hat{Y} = \beta0 + \beta1(\text{Square Footage}) + \beta2(\text{Number of Occupants}) + \beta3(\text{Appliances Used}) + \beta4(\text{Average Temperature}) + \beta5(\text{Building Type}) + \beta6(\text{Day of Week}) + \beta7(\text{Square Footage} \times \text{Average Temperature}) + \beta8(\text{Average Temperature} \times \text{Building Type})$

b. **Figure 20 –** Regression equations of Model 3

ii. Analysis of Variance of Model 3:

1. In the Analysis of Variance, we use the p-value 0.05 as criteria to reject the hypothesis that the model is not significant. Since the p-value of the model was of 0.000, we reject the hypothesis, therefore Model 3 is significant.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	**9**	**57312494**	**6368055**	**11017.58**	**0.000**
Square Footage	1	3017149	3017149	5220.07	0.000
Number of Occupants	1	7186329	7186329	12433.30	0.000
Appliances Used	1	5515108	5515108	9541.87	0.000
Average Temperature	1	22990	22990	39.78	0.000
Building Type	2	1448854	724427	1253.35	0.000
Square Footage*Average Temperature	1	2358	2358	4.08	0.047
Average Temperature*Building Type	2	4543	2271	3.93	0.024
Error	**70**	**40459**	**578**		
Total	**79**	**57352954**			

2. **Figure 21 –** Analysis of Variance of Model 3

i. T – Tests for Individual Regressor Significance of Model 3:

1. For all the regressors, we use the p-value 0.05 as criteria to reject the

hypothesis that the regressor is not significant. Since the p-value for all the regressor was of 0.000, we reject the hypothesis for all the regressors, meaning that all the regressors in Model 3 are significant

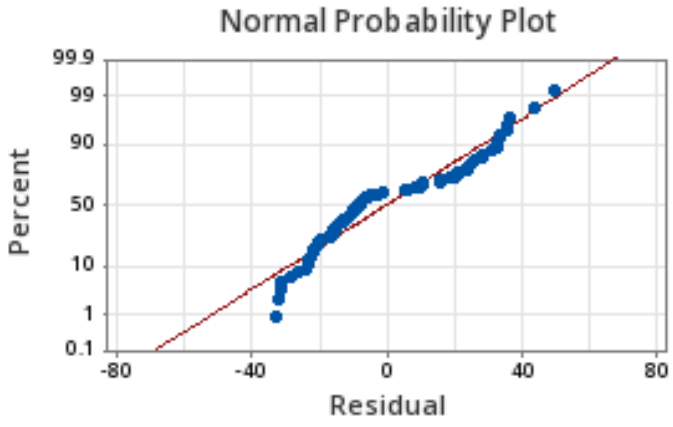
Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	**2069.7**	**30.5**	**67.95**	**0.000**	
Square Footage	**0.048986**	**0.000678**	**72.25**	**0.000**	**11.63**
Number of Occupants	**10.0253**	**0.0899**	**111.50**	**0.000**	**1.02**
Appliances Used	**20.336**	**0.208**	**97.68**	**0.000**	**1.13**
Average Temperature	**-7.98**	**1.27**	**-6.31**	**0.000**	**11.06**
Building Type					
Industrial	444.0	24.0	18.48	0.000	17.17
Residential	-566.8	24.6	-23.02	0.000	20.33
Square Footage*Average Temperature	**0.000059**	**0.000029**	**2.02**	**0.047**	**14.61**
Average Temperature*Building Type					
Industrial	2.45	1.07	2.30	0.025	18.49
Residential	2.97	1.09	2.74	0.008	23.00

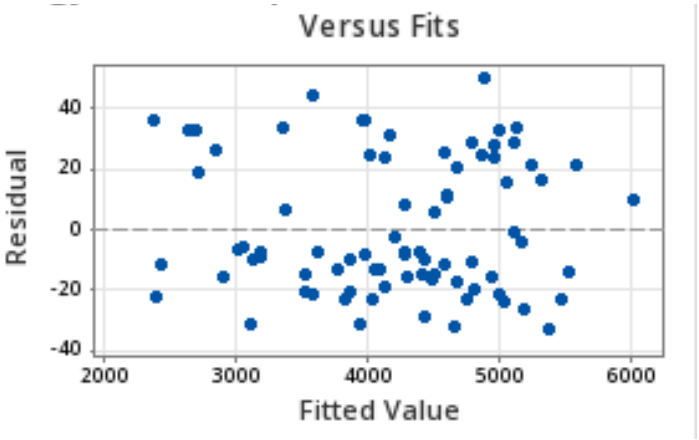
2. **Figure 22 –** T – Test for Model 3

i. Residual Analysis of Model 3 (Including plots and discussions):

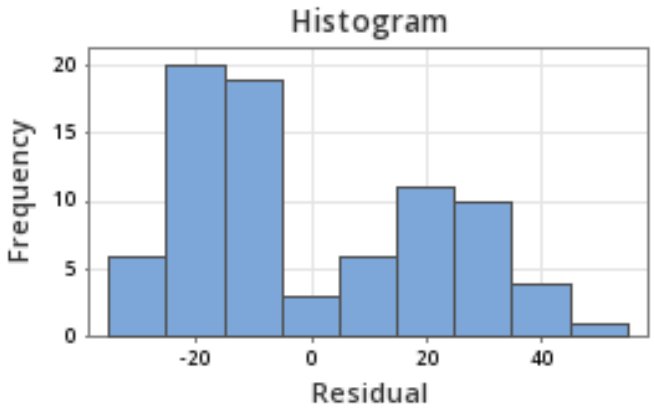
1. In Model 3, the residuals in the Normal Probability Plot appear to be close to the regression, meaning that they're close to following a Normal Distribution; and in the Versus Fits Plot, the data appears random. In the Histogram, the data is still segmented into two groups, however the data is connected unlike Model 2 and both groups appear to follow a Normal Distribution.



2. **Figure 23** – Normal Probability Plot for Model 3



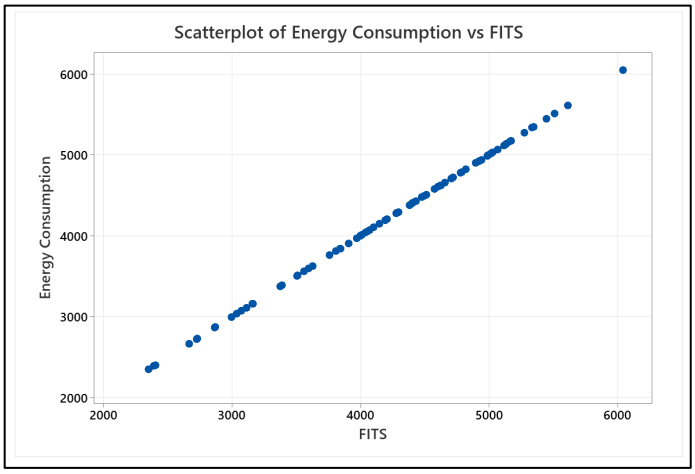
3. **Figure 24** – Scatterplot of Fitted Values vs. Fits for Model 3



4. **Figure 25** – Histogram of Frequency of Residuals for Model 3

d. Regression Model 4:

i. Regression Model Fitting of Model 4:



1. **Figure 26** – Regression Model Fitting of Model 4

ii. Regression Model Equation of Model 4:

1. The Regression Equation for Model 4 is:

   a. $\hat{Y} = \beta_0 + \beta_1$ (Square Footage) + β_2 (Number of Occupants) + β_3 (Appliances Used) + β_4 (Average Temperature) + β_5 (Building Type) + β_6 (Day of Week)

Regression Equation

Building Type	Day of Week			
Commercial	Weekday	Energy Consumption	=	2050.02 + 0.0500000 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 5.00021 Average Temperature
Commercial	Weekend	Energy Consumption	=	2000.01 + 0.0500000 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 5.00021 Average Temperature
Industrial	Weekday	Energy Consumption	=	2550.01 + 0.0500000 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 5.00021 Average Temperature
Industrial	Weekend	Energy Consumption	=	2500.01 + 0.0500000 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 5.00021 Average Temperature
Residential	Weekday	Energy Consumption	=	1550.02 + 0.0500000 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 5.00021 Average Temperature
Residential	Weekend	Energy Consumption	=	1500.02 + 0.0500000 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 5.00021 Average Temperature

2. **Figure 27** - Model 4 Regression Equation

iii. Analysis of Variance of Model 4:

1. In the analysis shown on Image #? we observe that every p-value is very small, displayed as 0.000, which indicates they are less than 0.001. This provides strong evidence that all predictors are highly significant.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	7	57352954	8193279	4.28927E+10	0.000
Square Footage	1	31286505	31286505	1.63788E+11	0.000
Number of Occupants	1	7196906	7196906	3.76765E+10	0.000
Appliances Used	1	5300030	5300030	2.77462E+10	0.000
Average Temperature	1	93266	93266	4.88255E+08	0.000
Building Type	2	13705991	6852995	3.58761E+10	0.000
Day of Week	1	45761	45761	2.39566E+08	0.000
Error	72	0	0		
Total	79	57352954			

2. **Figure 28** - Analysis of Variance (ANOVA) for Model 4

iv. T – Tests for Individual Regressor Significance of Model 4:

1. All VIF values are below 5, confirming that there are no multicollinearity issues. The predictor variables are sufficiently independent, ensuring stable and reliable coefficient estimates. The addition of Day of Week does not introduce multicollinearity problems.

Coefficients

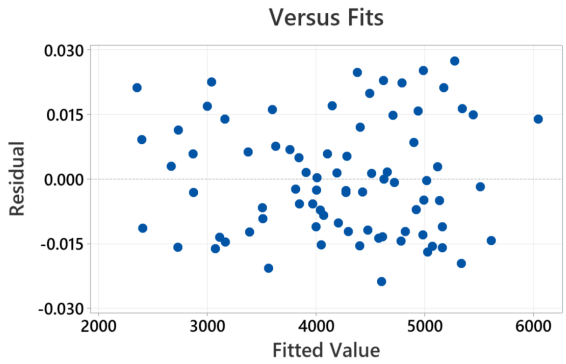
Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	2050.02	0.01	209544.52	0.000	
Square Footage	0.050000	0.000000	404707.42	0.000	1.17
Number of Occupants	9.9999	0.0001	194104.48	0.000	1.01
Appliances Used	19.9997	0.0001	166571.93	0.000	1.14
Average Temperature	-5.00021	0.00023	-22096.50	0.000	1.07
Building Type					
Industrial	499.997	0.004	118037.20	0.000	1.61
Residential	-499.993	0.004	-126020.47	0.000	1.60
Day of Week					
Weekend	-50.0061	0.0032	-15477.91	0.000	1.07

2. **Figure 29 –** T-Test for Model 4

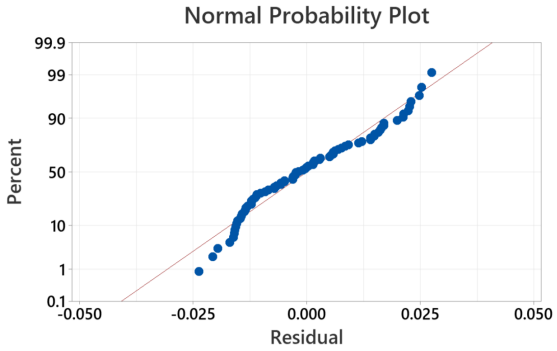
v. Residual Analysis of Model 4 (Including plots and discussions):

1. In the normal probability plot, we can observe that the residuals closely follow the normal line, with slight deviations at both extremes. This pattern indicates that the normality assumption is quite satisfied. In the Residual Versus Fits Plot, we can observe a random scattering of points around the horizontal line. The vertical spread remains reasonably consistent across all fitted values, indicating that the errors have

approximately a constant variance and are uncorrelated.



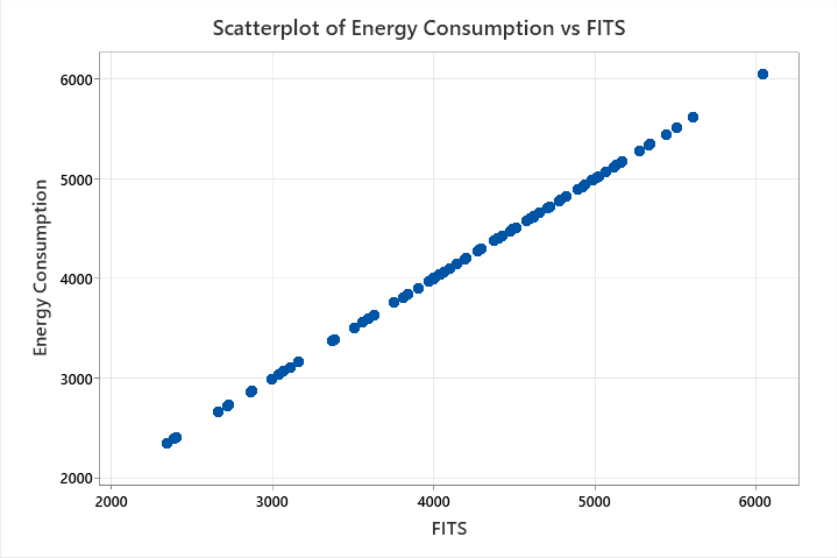
2. **Figure 30** – Scatterplot of Fitted Values vs. Fits of Model 4



3. **Figure 31 – Normal Probability Plot of Model** 4

e. Regression Model 5:

i. Regression Model Fitting of Model 5:



1. **Figure 32** – Regression Model Fitting of Model 5

ii. Regression Equation of Model 5:

1. $\hat{Y} = \beta 0 + \beta 1(\text{Square Footage}) + \beta 2(\text{Number of Occupants}) + \beta 3(\text{Appliances Used}) + \beta 4(\text{Average Temperature}) + \beta 5(\text{Building Type}) + \beta 6(\text{Day of Week}) + \beta 7(\text{Square Footage} \times \text{Average Temperature}) + \beta 8(\text{Average Temperature} \times \text{Building Type})$

Regression Equation

Building Type	Day of Week			
Commercial	Weekday	Energy Consumption	=	2050.00 + 0.050001 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 4.99970 Average Temperature - 0.000000 Square Footage*Average Temperatur e
Commercial	Weekend	Energy Consumption	=	2000.00 + 0.050001 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 4.99970 Average Temperature - 0.000000 Square Footage*Average Temperatur e
Industrial	Weekday	Energy Consumption	=	2550.00 + 0.050001 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 4.99949 Average Temperature - 0.000000 Square Footage*Average Temperatur e
Industrial	Weekend	Energy Consumption	=	2499.99 + 0.050001 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 4.99949 Average Temperature - 0.000000 Square Footage*Average Temperatur e
Residential	Weekday	Energy Consumption	=	1550.01 + 0.050001 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 4.99945 Average Temperature - 0.000000 Square Footage*Average Temperatur e
Residential	Weekend	Energy Consumption	=	1500.00 + 0.050001 Square Footage + 9.9999 Number of Occupants + 19.9997 Appliances Used - 4.99945 Average Temperature - 0.000000 Square Footage*Average Temperatur e

2. **Figure 33** - Model 5 Regression Equation

iii. Analysis of Variance of Model 5:

1. In the analysis shown in Image #__, the regression model is highly significant, with an overall p-value of 0.000. This indicates that the set of predictors included in Model 4 provides strong statistical evidence in explaining Energy Consumption. All main-effect predictors—Square Footage, Number of Occupants, Appliances Used, Average Temperature, Building Type, and Day of Week—have extremely small p-values (reported as 0.000), meaning $p < 0.001$. This confirms that each of these variables contributes significantly to the model. The interaction terms included in Model 4 (Square Footage × Average Temperature and Average Temperature × Building Type) have p-values of 0.102 and 0.926, respectively, which means they are not statistically significant.

Although these interaction effects do not add explanatory power, their non-significance does not affect the overall validity of the model. Overall, the ANOVA results show that Model 4 is statistically strong and explains nearly all the variability in Energy Consumption.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	10	57352954	5735295	3.04410E+10	0.000
Square Footage	1	3045999	3045999	1.61671E+10	0.000
Number of Occupants	1	7141844	7141844	3.79065E+10	0.000
Appliances Used	1	5142871	5142871	2.72966E+10	0.000
Average Temperature	1	8362	8362	44384844.70	0.000
Building Type	2	1415331	707666	3.75605E+09	0.000
Day of Week	1	40459	40459	2.14744E+08	0.000
Square Footage*Average Temperature	1	0	0	2.75	0.102
Average Temperature*Building Type	2	0	0	0.08	0.926
Error	69	0	0		
Total	79	57352954			

2. **Figure 34** - Analysis of Variance (ANOVA) for Model 5

iv. T – Tests for Individual Regressor Significance of Model 5:

1. In Image #__, we observe that some predictors have high VIF values, especially those related to Building Type and Average Temperature (ranging from 18 to 25). This indicates the presence of some multicollinearity among variables.

   However, all main predictors remain highly significant, and the model retains an excellent fit (R^2 = 100%, as shown in Image #__). Therefore, although moderate multicollinearity exists, it does not affect the stability or the overall interpretation of the model.

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	2050.00	0.02	117524.71	0.000	
Square Footage	0.050001	0.000000	127150.02	0.000	12.00
Number of Occupants	9.9999	0.0001	194695.87	0.000	1.02
Appliances Used	19.9997	0.0001	165216.86	0.000	1.17
Average Temperature	-4.99970	0.00075	-6662.20	0.000	11.93
Building Type					
Industrial	499.993	0.014	35107.67	0.000	18.50
Residential	-499.998	0.015	-33835.87	0.000	22.47
Day of Week					
Weekend	-50.0065	0.0034	-14654.16	0.000	1.21
Square Footage*Average Temperature	-0.000000	0.000000	-1.66	0.102	15.47
Average Temperature*Building Type					
Industrial	0.000215	0.000632	0.34	0.734	19.88
Residential	0.000247	0.000652	0.38	0.706	25.45

2. **Figure 35–** T-Test for Model 5

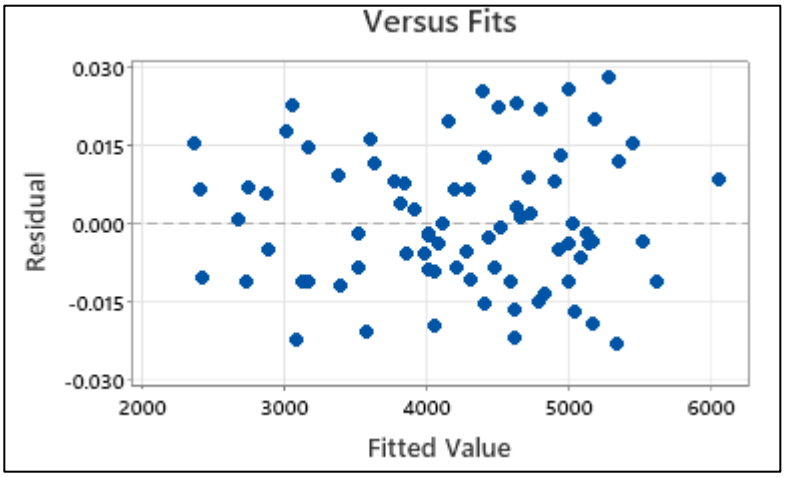
Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.0137261	100.00%	100.00%	100.00%

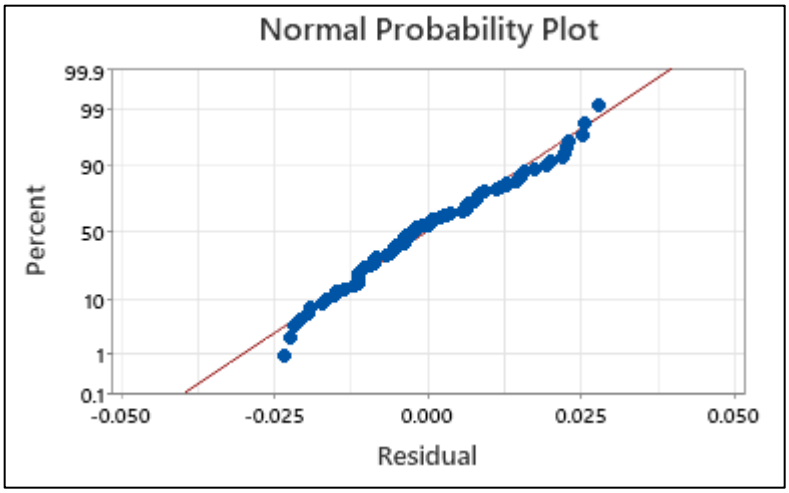
3. **Figure 36 –** Model Summary

v. Residual Analysis of Model 5 (Including plots and discussions):

1. In the Normal Probability Plot, the residuals align closely with the reference line, showing only minor deviations at the extreme ends. This indicates that the normality assumption is well satisfied, and the error terms follow an approximately normal distribution. In the Residuals versus Fits plot, the points appear randomly dispersed around the zero line with no visible patterns, curves, or increasing spread. This randomness confirms that the model meets the assumptions of constant variance and no correlation between errors. Overall, the residual diagnostics suggest that Model 4 satisfies the key regression assumptions and is statistically reliable.



2. **Figure 37-** Scatterplot of Fitted Values vs. Fits of Model 5



3. **Figure 38 -** Normal Probability Plot of Model

VI. Model Comparison and Selection:

Model	Predictors Included	R^2	*Adjusted* R^2	*Predicted* R^2	Standard Error	MAPE
1	Square Footage, Occupants, Appliances, Building Type	99.79%	99.77%	99.75%	40.50 kWh	0.79%
2	Square Footage, Occupants, Appliances, Building Type, Average Temperature	99.92%	99.91%	99.90%	25.037 kWh	0.51%
3	Square Footage, Occupants, Appliances, Building Type, Average Temperature, Temperature * Square Footage	99.93%	99.92%	99.91%	24.041 kWh	0.56%
4	Square Footage, Occupants, Appliances, Building Type, Average Temperature, Day of the Week	100%	100%	100%	0.0138 kWh	0.00%
5	Square Footage, Occupants, Appliances, Building Type, Average Temperature, Day of the Week, Square Footage * Average Temperature, Average Temperature * Building Type	100%	100%	100%	0.0137 kWh	0.00%

1. **Figure 39 –** Comparative Table of Training and Testing Metrics Across all Five Models

ii. Figure 39 presents the diagnostic metrics for all five models, which will be ranked by predictive precision and explanatory power. Model 1's performance was very limited, as it included only four predictors: Square Footage, Number of Occupants, Appliances Used, and Building Type. In contrast, the final model, Model 5, incorporated all available variables plus two interactions: Square Footage x Average Temperature and Average Temperature x Building Type. While the visual difference in Adjusted R^2 between the two models appears negligible, the Standard Error reveals a significant contrast. Model 5 is more precise than Model 1, with a Standard Error of 0.01 kWh compared to Model 1's 40.5 kWh. Model 1 acted as a baseline; while good for general estimation, it failed to capture the building dynamics and the specific environment of each observation. Conversely, Models 2 and 3 accounted for these environmental factors by including Average Temperature as a predictor. Finally, Models 4 and 5 captured the behavioral interactions of the building.

iii. Understanding the mathematical complexity of each model is essential for evaluating performance. In this case, adding complex mathematical terms yielded low returns. While adding a simple variable like Average Temperature in Model 2 diminished the Standard Error by 15 kWh, adding the interaction term (Temperature x Square Footage) in Model 3 only reduced it by 1kWh. Despite Model 3 outperforming Model 2, Model 3 was chosen as the least efficient model because the added complexity was negligible and did not add any predictive power. This hierarchy illustrates a natural transition in the behavior of each model. The final models, Models 4 and 5, incorporate real-world interactions like those between Average Temperature and Building Type.

iv. Although the interaction terms in Model 3 were statistically significant (P-value lower than 0.05), they lacked practical significance in the evaluation of the models. A reduction of 1 kWh is operationally invisible, and hence, not a factor that should be considered. Predicting terms should not be evaluated nor included based on P-values alone. They should be significantly improving the model's predictive power to be considered. Apart from the issues mentioned before, Model 3 is not linear. Non-linearity is more complex to act upon for building managers, offering no value in contrast to a simpler model. Model 4 represents a clearer and actionable operational state; it considers human behavior based on the day of the week. Based on the Principle of Parsimony (Occam's Razor), if two explanations are adequate, the simpler one is usually better. Model 4 predicts energy consumption based on the weather, the physical load of each building, and the weekday human interactions.

v. To validate the relationships in Model 4 and to obtain accurate predictive power, we assessed the data in Model 5. Figure 39 shows that none of the

models are capturing random noise nor missing significant trends, given by the minimal difference between Adjusted R^2 and Predicted R^2. Finally, Model 5 is selected as the final model due to its predictive power and statistical significance, distinguished by its negligible Mean Absolute Percentage Error (MAPE) of 0.0% and its Standard Error of 0.0137 kWh. While Model 4 is more parsimonious, Model 5 confirms that the variables used are the optimal combination of the physical factors of the building, the average temperature of the environment, and the human factors driven by the day of the week. While Model 4 is more parsimonious, Model 5 provides the mathematical ceiling of the dataset. Manually including interaction terms allowed us to assess the significance of each predicting term, which was low due to high P-values in the model, and to ensure all potential variances were explained.

VII. Discussion and Recommendations:

a. To assess the practical application of Model 5, the regression equation was applied to three hypothetical scenarios to validate its predictive precision. For a typical Commercial office operating on a weekday (25,000 sq. ft., 50 occupants, 30 appliances, 20°C), the model predicted a daily consumption of 4,300 kWh. In a smaller Industrial scenario operating on a weekend (15,000 sq. ft., 10 occupants, 50 appliances, 15°C), the model yielded a prediction of 4,274.99 kWh, reflecting the high base load inherent to industrial structures despite the smaller footprint. Finally, for a large Residential complex on a weekday (40,000 sq. ft., 100 occupants, 20 appliances, 25°C) the model estimated 4,825.01 kWh. These calculations confirm that Model 5 can provide Lux Inc. with precise consumption estimates based on structural and operational variables.

b. This regression analysis demonstrated that Square Footage is the dominant predictor of energy consumption since it consistently accounts for a large portion of the variability across all five models. However, analyzing Building Type proved equally critical. Industrial buildings possess a significantly higher baseline energy demand than any other building type regardless of size. Furthermore, the Appliances Used variable showcased high impact, adding approximately 20 kWh of consumption per unit; Day of the Week confirmed that energy usage dropped on the weekends. Although Average Temperature showed a statistically significant negative relationship, its overall influence was smaller when compared to other factors (structural components and appliances).

c. Lux Inc. should adopt a targeted approach to improve its energy efficiency. Prioritizing energy consumption audits for Industrial clients could offer the highest energy savings. Given the 20-kWh impact per appliance, Lux Inc. should develop a program that encourages costumers to seek for energy-efficient equipment. This simple initiative could provide a tangible energy consumption reduction. For operational planning, the

company can utilize temperature and weekday coefficients to estimate grid planning and capacity during cold fronts or peak weekdays. Finally, while Model 5 provides a mathematical ceiling with 100% accuracy, we recommend Model 4 for general field use by Lux Inc. representatives. Abiding by Occam's Razor, Model 4 offers near-perfect predictive power without the complexity of interaction terms.

VIII. Conclusion:

IX. References:

a. YouTube. (n.d.). YouTube. https://www.youtube.com/watch?v=Ie53YZiRZy0

X. Appendix